

Insights and Trends in Diabetes Education

Results of the 2008 AADE National Diabetes Education Practice Survey

Annette Lenzi Martin

Terry Lumber, RN, CNS, CDE, BC-ADM

Terry Compton, MS, APRN, RN, CDE

Kristina Ernst, BSN, RN, CDE

Linda Haas, PHC, RN, CDE

Janet Regan-Klich, EDS, RD, CDE, FADA

Nancy Letassy, PharmD, CDE

Karen A. McKnight, RD, LD

Joseph B. Nelson, MA, LP

Jane Jeffrie Seley, MPH, MSN, GNP, CDE

Judith A. Toth, PharmD, CDE, CGP, FASCP

Carole Mensing, RN, MA, CDE

From Lenzi Martin Communications, Inc, Chicago, Illinois (Ms Martin); Inova Diabetes Center, Fairfax, Virginia (Ms Lumber); Southeastern University School of Nursing, Hammond, Louisiana (Ms Compton); Division of Diabetes Translation, Atlanta, Georgia (Ms Ernst); Endocrinology Clinical Specialist, VA Puget Sound Seattle Division, Seattle, Washington (Ms Haas); University of Chicago, Chicago, Illinois (Ms Regan-Klich); University of Oklahoma College of Pharmacy, Edmond, Oklahoma (Ms Letassy); St Joseph Healthcare Diabetes Treatment Center, Lexington, Kentucky (Ms McKnight); Behavioral Medicine, Gordon Valley, Minnesota (Mr Nelson); New York Presbyterian/WC, New York, New York (Ms Seley); Jesse Brown VA Medical Center, Chicago, Illinois (Ms Toth); Joslin Diabetes Center, Boston, Massachusetts (Ms Mensing).

Correspondence to Terry Lumber, RN, CNS, CDE, BC-ADM, Director, Inova Diabetes Center, 2700 Prosperity Ave, Suite 100, Fairfax, VA 22301-4330 (terry.lumber@inova.org).

Purpose

The purpose of this study is to describe current diabetes education practice and specific interventions and responsibilities of diabetes educators in the United States.

Methods

The 2008 National Practice Survey (NPS) instrument consisted of 53 items addressing diabetes education program structure, processes and interventions, outcomes and quality improvement activities, and the chronic care model. The survey was hosted online for American Association of Diabetes Educators (AADE) members. Participants totaled 2447 members, constituting a 25% return rate. Data from the 2008 NPS were analyzed and compared with results from previous surveys.

Results

Nearly two-thirds of respondents in 2008 provided diabetes education in a single location, most commonly in a clinical outpatient/managed care setting (39%). Most programs provided comprehensive services. Managers noted that 42% of their programs were either cost/revenue neutral or profitable. Programs varied in types of services, number of patient visits, team member functions, time spent on services, and instructional methods used. At least 50% of managers said their programs report outcome data, and 88% participate in quality/performance improvement activities. Nearly two-thirds of respondents were unfamiliar with the AADE-adopted chronic care model.

DOI: 10.1177/0145721708327286

Conclusions

Many 2008 NPS results concur with those obtained in 2005 through 2007. Areas of variability among programs suggest a need for standardized interventions and practice guidelines. Educators are encouraged to report outcomes to elucidate the contributions of their programs to patient care. AADE can use the results and comparative data obtained from the 2008 survey when developing practice, research, and advocacy activities.

As the health care landscape continually evolves and the prevalence of diabetes remains a serious concern, diabetes educators in the United States face both challenges and opportunities. One of the goals of the American Association of Diabetes Educators (AADE) is to set the scope and direction for the practice of diabetes education, and to do so, it is incumbent upon the association to take the pulse of its members and procure an accurate description of the programs they provide. Toward that end, AADE has been gathering and evaluating survey information addressing the practice of diabetes education in the United States.

A major starting point in collecting such data entailed an 18-month environmental scan of AADE members and the external health care environment; this has been followed by more extensive national surveys on diabetes education practice. Prior to these assessments, diabetes education surveys consisted largely of job analyses that addressed demographics and tasks performed, but they did not describe the practice setting or specific interventions of diabetes educators.¹

The environmental scan, completed in 2005, yielded qualitative data that revealed key trends affecting the field of diabetes education.² As valuable as the environmental scan was, the need for a more descriptive analysis of diabetes education became apparent.³ Serving as a springboard for future research, data from the environmental scan were used to help develop a national practice survey. What emerged was the AADE National Diabetes Education Practice Survey (NPS), administered each year from 2005 to 2008. The first NPS was undertaken to obtain quantitative and qualitative data to more accurately describe diabetes education practice and form the baseline for monitoring changes. Results from the 2005 and 2006 surveys have been previously reported.²

This article summarizes the major findings of the 2008 NPS, highlighting trends and significant departures from the results of previous years. The purpose of the AADE NPS has remained the same since its inception: to describe the current diabetes education practice along with specific interventions and responsibilities of diabetes educators. As with previous NPS findings, the 2008 results and comparative data will provide guidance and input for AADE in the development and fine-tuning of professional practice, research, and advocacy activities.

Methods

Survey Instrument

The AADE National Diabetes Education Practice Survey incorporates the Donebedian model, which consists of structure (ie, the organization of the system's components), process (ie, the work of the system), and outcome (ie, the result of the interaction between structure and process).^{2,4} In the NPS, a diabetes education program is defined as "any structured, organized delivery of education occurring in any practice setting." For each year's survey, AADE has retained the services of Perception Solutions, Inc (Aurora, IL), a market research company specializing in assisting associations with the graphic design, Web enablement, and analysis of results of membership surveys. All matters pertaining to the NPS fall within the purview of the AADE Professional Practice Committee.

Since the launch of the NPS in 2005, the survey instrument has undergone minor formatting and content refinements, some of which were based on feedback received through respondents' comments. In 2007, the survey added 3 questions on the chronic care model (CCM) and 2 questions on the AADE 7TM Self-Care Behaviors framework, thus addressing these constructs for the first time in the NPS. Changes in the 2008 survey involved relocation of some items and inclusion of a skip pattern that permitted only diabetes program managers to answer many of the questions.

A total of 53 questions appeared in the 2008 survey, 4 of which were open-ended items eliciting comments on the survey's first 3 sections and on the survey as a whole. Questions were organized into 4 sections, as follows:

1. Diabetes education program structure (29 questions)
2. Diabetes program process and interventions (10 questions)
3. Diabetes program outcomes and quality improvement (8 questions)
4. CCM (5 questions)

Survey Distribution

Approximately 9900 AADE members were sent an electronic invitation to participate in the survey, with 564 e-mails returned as undeliverable. The 2008 NPS was hosted online in June. A total of 2447 members responded to the survey, resulting in a statistically significant 26% return rate. This return rate is consistent with that of prior years' surveys.

Analysis

The results of the 2008 NPS were analyzed by Perception Solutions, Inc. Based on the sample size of several hundred for most questions, they determined there was a 95% confidence level that the survey results are within $\pm 5\%$ accuracy of the stated values or better. A trend analysis also was conducted comparing 2008 results with those from previous years to assess for significant changes. Several items showed statistically significant differences, as determined at the significance level of .01 or lower.

Results

Of the 2447 AADE members who participated in the survey, 1332 (60%) indicated their primary role was "diabetes educator," and 435 (20%) selected "diabetes program manager/director/coordinator" as their primary role. All participants were asked to respond to the first 10 questions, which elicited primarily demographic information. Those who indicated "diabetes educator" or "other" in question 11 were instructed to skip to question 48 in the survey's last section (Chronic Care Model). Thus, most items (questions 12-47) were answered solely by diabetes program managers, directors, or coordinators (referred to as "program managers" or "managers" in this article).

Characteristics of Respondents

Most survey participants identified their disciplines as either "registered nurse" (40%) or "registered dietitian" (30%). About 92% of all respondents reported that they were "providing, supervising, or coordinating diabetes patient education" at the time they completed the survey, and nearly 62% were full-time educators. Additional characteristics of survey participants are shown in Table 1.

Settings for Diabetes Education Programs

Most employed respondents provided diabetes education at 1 location (64%) in 2008, while about 36%

Table 1

Characteristics of the AADE National Practice Survey 2008 Survey Participants

Characteristic	%
Respondents (return rate), n (%)	2447 (26.2)
Primary role	
Diabetes educator	60.0
Diabetes program manager/director/ coordinator	19.6
Other	20.4
Discipline	
Registered nurse	40.2
Registered dietitian	29.7
Program coordinator	7.9
Nurse practitioner	6.4
Clinical nurse specialist	6.3
Pharmacist	3.3
Psychologist/behavior specialist/ social worker	0.5
Researcher, data manager, etc	0.5
Exercise physiologist/physical therapist	0.4
MD, endocrinologist	0.1
Other	4.5
Credentialing	
Certified diabetes educator	79.7
Board-certified advanced diabetes management	5.2

worked at multiple sites—mostly 2, 3, or more than 10. The top 5 states where diabetes education programs were located were California, Texas, New York, Pennsylvania, and Ohio.

Survey participants were given 8 choices (including "other") to indicate their primary setting for providing diabetes education. The most common setting was clinical outpatient/managed care (39%). Other responses included clinical inpatient (16%), community-based setting (13%), other (12%), and private practice/physician (11%). When asked to describe their current employment, respondents were fairly equally split among community-based organization (16%), educational institution/academia (13%), diabetes-related company (13%), and government

organization (10%). However, the greatest proportion of respondents indicated “other” (38%).

Patient Populations

Questions regarding the populations served by diabetes education programs were answered only by program managers. The most common patient age groups were 45 to 64 years old (41%) and 65 years or older (31%), while 21% were 19 to 44 years old and 5% were younger than 19 years. Most of the patients (73%) had type 2 diabetes. Far less common diagnoses were type 1 diabetes (10%), gestational diabetes (7%), and prediabetes (5%), defined as impaired glucose tolerance or impaired fasting glucose. A sizable proportion of patients (43%) were newly diagnosed (ie, <6 months) in 2008.

About 61% of patients were Caucasian/white, 17% were Black/African American, and 8% were identified as other/mixed. Asian patients and American Indian/Alaska Native patients each accounted for 3% of all patients, and Native Hawaiians and Pacific Islanders comprised less than 1%. No choice was provided in the 2008 survey for Hispanic, which in previous years accounted for about 10% of the patient population. English was the primary language for 88% of patients. Spanish was the first language of 8% of patients.

About one-third of patients had a high school or GED degree as their highest education level (37%), and 11% were noted as having some high school education. In terms of advanced education, 10% of patients had acquired some college education, and 14% had received a college degree or postgraduate education. Six percent of patients either had no formal education or had gone no further than elementary school. More than one-fifth of managers (22%) said they did not know the educational level of their target audience.

Administration/Operations of Diabetes Education Programs

Most of the program managers (70%) reported that their diabetes education program has been in operation for more than 5 years. Approximately two-thirds of programs (68%) are recognized by the American Diabetes Association (ADA) Education Recognition Program, whereas 16% of programs were free-standing and not recognized in 2008.

Completion of diabetes education programs was achieved by 68% of patients in 2008. When asked how many patient visits their program provided in a year, managers were evenly split among 3 categories, ranging

from 201 to 2000 patient visits (see Figure 1). Typical program services, such as those relating to type 1, type 2, and pediatric diabetes, averaged 8.5 hours and 4.6 visits (see Table 2). However, much variability was reported in the time interval from start to finish of services, ranging from 1 to 2 weeks to 17 to 24 weeks. The primary staff involved in diabetes education in 2008 represented a variety of disciplines (see Table 3).

Nearly one-third (30%) of diabetes education programs were reimbursed by Medicare in 2008. Managed care accounted for the reimbursement of 17% of programs, private insurance companies for 14%, and Medicaid for 9%. In describing the financial status of diabetes education programs, 42% of managers said their programs operated at a loss. About one-third indicated their programs were revenue/cost neutral (34%), while only 10% of managers reported operating at a profit.

Services Provided in Diabetes Education Programs

Most diabetes education in 2008 consisted of comprehensive programs. About 72% of managers indicated that their program was comprehensive, comprising initial education and education that includes assessment, goal setting, intervention, and evaluation. In contrast, 26% of programs were reported to be follow-up education (ie, topic-specific, focused teaching that is not part of a comprehensive program). When asked to indicate the primary setting for providing instruction/education, most managers (66%) selected “1 to 1 and group,” 19% selected “group,” and 14% said “1 to 1 (individual).” Telemedicine was reported by only 2 managers (0.6%).

The types of professional services provided were shown to vary among programs, with managers most frequently selecting diabetes self-management education/training (DSME/T; 76%) and medical nutrition therapy (60%). These were followed by clinical (medical management; 37%), health care professional education (35%), disease management (31%), telephone care management (29%), case management (24%), counseling (23%), activities unrelated to diabetes (14%), research (9%), home-care diabetes education (8%), and other (4%).

Diabetes education programs also offered a wide range of patient support services in 2008. Nearly one-half of programs provided telephone follow-up services (47%) and support groups (46%). Other patient services focused on weight loss (39%), foot examinations (36%),

Table 2

2008 Program Services: Time Devoted and Number of Visits

Program/Service	Service Provided (% Selected)	Total Hours to Complete Entire Program (Average)	Total Number of Visits to Complete Entire Program
General diabetes	40	8.2	4.0
Type 2 diabetes	46	8.9	4.3
Type 1 diabetes	38	9.0	4.6
Gestational diabetes	37	4.1	3.0
Insulin initiation/adjustment	39	2.9	2.5
Insulin intensification	22	3.1	2.8
Medical nutrition therapy	40	3.2	2.7
Insulin pump initiation	25	6.8	3.8
Continuous glucose monitoring system	16	2.7	2.1
Case management	7	6.0	5.6
Pediatric	8	7.6	4.9
Other	2	3.1	2.6

Table 3

Primary Staff Involved in Diabetes Education in 2008

	Average Weekly Hours Devoted to Diabetes Education		
	Number of FTEs	DSME/T	Clinical Management
Community/lay health worker	3.56	10.45	10.36
Exercise physiologist/physical therapist	0.64	7.23	3.88
Licensed practical nurse	0.57	7.82	5.98
MD, Primary care	0.51	5.75	9.74
MD—Endocrinologist	2.40	7.30	18.22
Nurse practitioner / Clinical nurse specialist	1.26	15.16	21.55
Pharmacist	1.95	8.03	12.63
Physician assistant	0.56	5.18	7.76
Program coordinator	1.30	17.80	15.14
Psychologist or behavioral specialist/social worker	0.66	7.88	8.38
Registered dietitian	2.26	31.91	17.17
Registered nurse	2.76	36.74	21.07
Support/clerical staff	1.46	16.65	9.87

CDE, certified diabetes educator; DSME/T, diabetes self-management education/training; FTE, full-time employee.

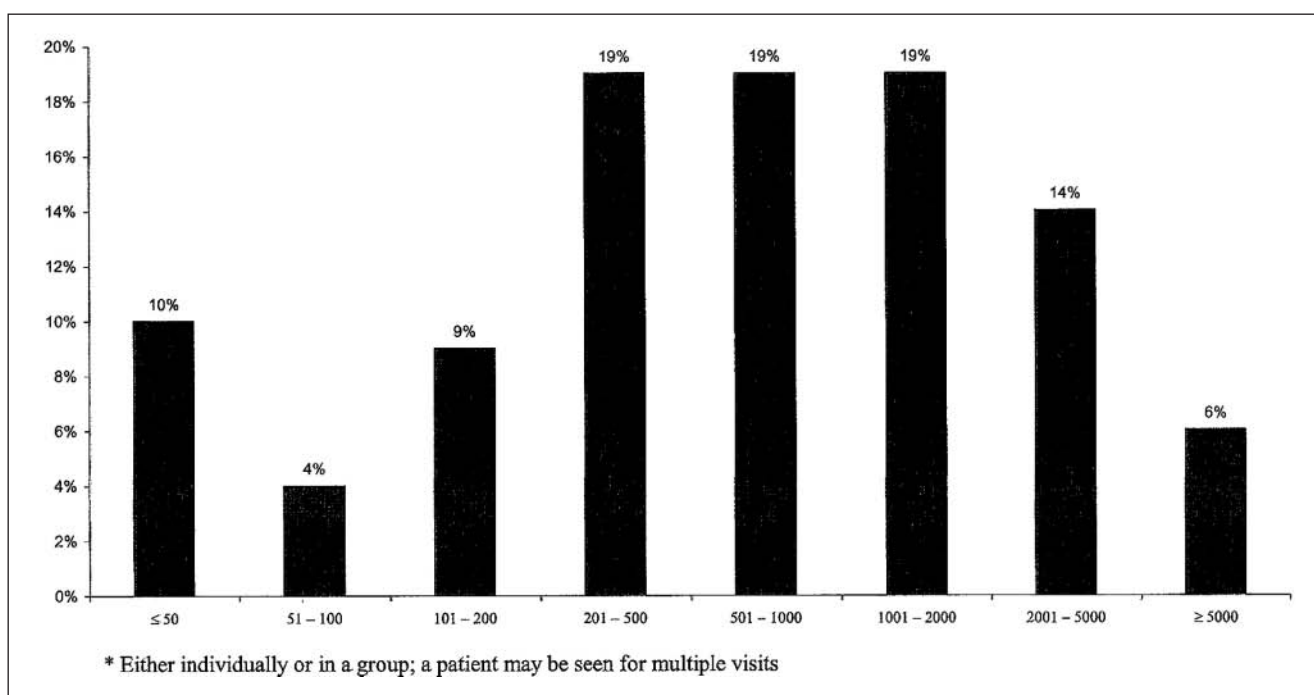


Figure 1. Total number of patient visits per year provided in diabetes education programs (either individually or in a group; a patient may be seen for multiple visits).

clinical (medical management; 34%), exercise training (27%), smoking cessation (25%), cholesterol/lipid screening (22%), flu shots and immunization (21%), stress management (21%), eating disorder counseling (13%), mental health counseling (13%), and eye examinations (10%).

Program Curricula, Methods, and Strategies

Most curricula for diabetes education programs reflected either purchased products that underwent staff revision (37%) or curricula designed solely by staff (35%). About 14% of curricula were developed by consultants. Most curricula were edited on an ongoing basis (63%), while 33% were edited annually.

Various methods were used in 2008 to deliver diabetes education. Print, audio, and visual materials (77%) and goal setting (74%) led the list. More than half of managers also reported implementing skills demonstration labs (56%). Figure 2 shows all the instructional/educational delivery methods reported by managers.

Managers also were asked to indicate the behavioral strategies they incorporated in their diabetes education. Situational problem solving (67%) was the most common, followed by patient contracts (44%) and cognitive

reframing (37%). Other behavioral strategies included conviction and confidence scaling (20%), relapse prevention training (15%), and stimulus control (8%).

Outcome Measurement and Quality/Performance Improvement

Several behavioral outcomes are addressed in diabetes education programs. About 51% of programs collect and report outcomes for each of the following: eating/nutrition, physical activity, and monitoring. Other behavioral outcomes that are reported focus on diabetes complications risk reduction (42%), medication administration (39%), problem solving (especially for high, low, and sick day blood glucose levels; 33%), and psychosocial adaptation (25%). Approximately 5% of managers selected “other,” and another 5% did not know.

In terms of learning outcomes that were collected and reported, 35% focused on skills acquisition and 33% on knowledge scores. About 9% of managers selected the option “other,” while 14% said their program does not collect any learning outcome data and another 14% selected the “don’t know” option.

Among the clinical outcomes collected and reported, hemoglobin A1C heads the list (57%), followed distantly

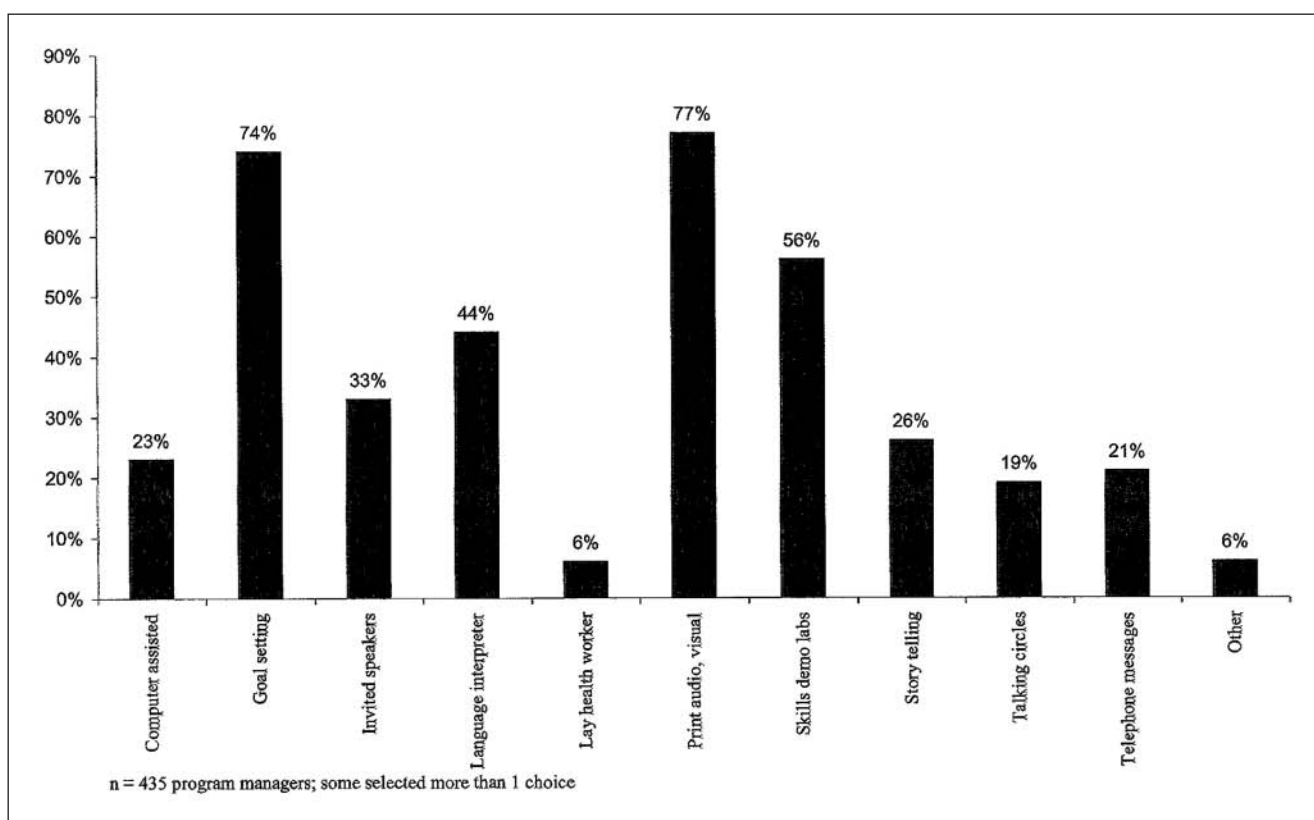


Figure 2. Instruction/education delivery methods used as reported by program managers (n = 435 program managers; some selected more than 1 choice).

by weight (33%). Only 24% of programs measure and report their patients' blood pressure. Other outcomes measured and reported include total cholesterol (21%), triglycerides (19%), body mass index (18%), microalbumin (16%), and "other" (10%). About one-third of programs collect and report data on foot examinations (35%) and eye examinations (32%), while 9% of managers said they collect no data on these process measures and 11% selected "don't know."

In 2008, the customers for most of the outcome reports were diabetes advisory groups (72%), followed by accreditation agencies such as the ADA (60%) and internal administration (43%). In 2 noticeable increases from previous years, 89% of managers said their program collects patient satisfaction surveys, and 88% participate in quality or performance improvement activities. The amount of time spent on data entry, outcomes collection, and reporting varied in 2008, with 52% of respondents selecting ranges from 3 to 10 hours per month.

Chronic Care Model

All survey participants were asked to answer the 5 items that comprised the Chronic Care Model section of the 2008 survey. These questions probed the awareness among members of 2 constructs espoused by the AADE for the provision of diabetes education and care: the AADE7TM Self-Care Behaviors and the Chronic Care Model.

The AADE7TM Self-Care Behaviors offers a framework for patient-centered diabetes education and a view of where that education fits into the diabetes care continuum.⁵ Nearly all respondents (96%) said they were familiar with these behaviors. However, not all AADE members are taking advantage of tools that can enhance their ability to help patients maintain the 7 behaviors. About 59% reported that they use AADE7TM Impact, a suite of Internet-based tools designed to aid educators in such areas as behavior change tracking, report generation, and data collection. AADE7TM Impact, which is part of the AADE7TM System, is available to members at no charge.

Three survey questions explored the CCM created by Wagner of the MacColl Institute for Healthcare Innovation. This model, adopted by the AADE in 2007, is considered an ideal framework to support diabetes self-management education.⁶ Nearly one-half of respondents (49%) said they were interacting with providers who use a CCM, but only 30% were familiar with the MacColl Institute's CCM (also called the Wagner chronic care model). Among those who knew of this model, 75% said they included it in self-management support. Other areas in which the CCM was being used were the community (40%), health system (36%), decision support (35%), delivery system design (28%), and clinical information systems (27%).

Discussion

The National Diabetes Education Practice Survey has served as a useful tool in gathering information from the AADE membership to describe the practice of diabetes education in the United States from 2005 to 2008. The return rate for the NPS has been statistically consistent over the years. The 26% return rate of the 2008 NPS (encompassing 2447 participating members) was slightly lower than the 30% return rates achieved in both 2006 (2709 respondents) and 2007 (3109 respondents), but it was higher than the 21% rate in 2006 (1781 respondents).

Eighty percent of 2008 respondents were either diabetes educators or diabetes program manager/director/coordinators; this was similar to previous years. As with past surveys, the discipline mix for the 2008 respondents is largely representative of the general membership of AADE, with most either registered nurses or registered dietitians. However, 80% of 2008 respondents reported having the CDE credential, in contrast to the 55% reported by the 2008 AADE general membership. One possible explanation for this disparity may be a greater tendency among CDEs to participate in information-gathering efforts such as surveys. More than half of diabetes education programs were provided primarily as either outpatient or inpatient services in 2008. Because the answer choices for primary setting differed substantially in previous years, a comparative assessment cannot be made.

Comparisons can be made, however, for most of the questions in the 2008 survey. Many 2008 findings were consistent with those from prior years, particularly regarding program structure. For example, diabetes education

programs have consistently been fairly evenly divided among urban, suburban, and rural locations. Furthermore, the percentage of newly diagnosed patients in programs has ranged from 43% to 46%, indicating that more than half of patients have had a diagnosis of diabetes for longer than 6 months. In each year since 2005, about two-thirds of patients complete their diabetes education.

Generally, the team members primarily involved in providing DSME/T in 2008 (Table 3) and previous years, as determined by average weekly hours spent, were identified as registered nurses, registered dietitians, program coordinators, and nurse practitioners/clinical nurse specialists. Staff involvement in clinical management was similar in all 4 years, with the exception of registered dietitians, who for the first time in 2008 were among the top 4 disciplines devoting the most time to clinically managing patients in diabetes education programs. NPS results have consistently revealed that team members perform a variety of program administration functions. Typically, the team includes a program manager and a complement of health care professionals.

Reimbursement of diabetes education has remained virtually the same since 2005. Because about 30% of patients consistently fall into the age group of 65 years or older, Medicare leads the list as a source of reimbursement, followed by managed care entities and private insurance companies.

The balance sheets of diabetes education programs also have shown consistency since 2005, indicating that on average, 42% of programs operate at a loss, 12% operate at a profit, and 31% break even in revenue/costs. Given the fact that many diabetes education programs are provided at no charge, the implications of these findings are unclear. Interestingly, 13% of program managers were unsure of the financial operating status of their program in 2008, similar to previous years' findings.

Trending Highlights

The results of the 2008 NPS were compared with those from prior years to identify significant differences in the practice of diabetes education. In the trend analysis, statistically significant differences were noted at the .01 level, indicating that the likelihood of a trend is 99%. Several items emerged as having statistically significant differences:

Number of locations and years of operation. A significant increase in the proportion of educators who

work at only 1 location was noted for 2008 compared with previous years. This contrasts sharply with a trend noted from 2005 through 2007, when working at multiple locations was on the rise. In 2008, 64% of educators served at a single site compared with 57% in 2007.

Diabetes education programs are beginning to experience longevity, as revealed by a trend occurring over the past 4 years. Compared with 2005 data, 2008 results show a significant increase in programs operating for more than 5 years (69% vs 58%).

Patient population characteristics. The racial distribution of patients receiving diabetes education has undergone 2 significant trends from 2006 to 2008. First, an increase has occurred in the Asian population, rising from 2.4% to 3.4%. Even more dramatic is the increase in the “other/mixed” response, growing from 1.3% to 8.4%. This upswing may reflect, at least in part, the omission of “Hispanic” from the choices in the 2008 survey.

A small but statistically significant change regarding patients’ primary language has been occurring since 2006. The selection of “other” has risen steadily—from 1.9% to 3.1%—indicating that neither English nor Spanish is the first language in a growing number of patients, and perhaps supporting the findings of increased Asian and “other/mixed” populations. In another small but statistically significant difference, the proportion of patients with no formal education has been declining since 2006.

Services provided and methods used. Since the first NPS was administered in 2005, comprehensive diabetes education has increasingly replaced topic-specific, focused teaching (follow-up services). The provision of follow-up services has dropped from 34% to 26%, constituting a statistically significant change.

Three significant declines occurred in 2008 compared with 2007 in professional services provided: disease management decreased from 41% to 31%, health care professional education from 42% to 35%, and research from 14% to 9%. However, the declines in disease management and research contradict the previous significant increases seen when comparing 2007 and 2006 results.

Print, audio, and visual materials consistently rank as the most common instructional delivery method used, but in a significant change from past surveys, more than half of the 2008 respondents (56%) said they now use skills demonstration labs, up from 46% in 2005. Another

striking change is the increased use of “talking circles” in 2008 (19%) versus 2007 (11%), overturning the decreased use of this method from 2006 to 2007. More programs also used a language interpreter in 2008 (44%) than in 2007 (37%), but this difference was not statistically significant.

Diabetes education programs continue to offer various patient support activities. Even though weight loss was among the top 3 patient support services reported in 2008, a statistically significant decline was noted, with 39% of programs offering weight loss services in 2008 compared with 48% and 43% in 2007 and 2006, respectively.

Program managers also have reported the declining use of 2 types of behavioral strategies. Since 2006, cognitive reframing has steadily decreased from 47% to 36%, reflecting a definite trend. In addition, statistically significant declines occurred in the use of stimulus control, dropping from 12% in 2007 to 8% in 2008.

Patient satisfaction remains a high priority among diabetes educators, and in 2008, a significant spike took place. Eighty-nine percent of programs collected patient satisfaction surveys in 2008, compared with an average 74% during previous years.

Curriculum development and maintenance. Although most diabetes education curricula continue to consist of either staff-revised purchased products or staff-designed programs, a significant downturn is noted regarding staff involvement in creating curricula from scratch. Whereas more than half (56%) of curricula were designed by staff in 2005, only about one-third (35%) were staff designed in 2008.

Outcomes collection and reporting. Trends, primarily downward, were revealed in the collection and reporting of outcomes by diabetes education programs.

For behavioral outcomes, more than one-half of programs collect and report data on eating/nutrition, physical activity, and monitoring. However, outcome activity regarding monitoring has decreased significantly since 2005, from 61% to 51%. In addition, medication administration outcome activity has dropped significantly every year since 2005, from 46% to 37%.

The collection of learning outcomes (knowledge scores and skill acquisition) increased from 2007 and 2008 but not significantly. Two statistically significant differences did occur from 2005 to 2008, but their mixed messages make it difficult to assess this area. On one

hand, each year fewer respondents answered “none collected” (decreasing from 22% to 14%). On the other hand, each year more respondents answered “don’t know” (growing from 8% in 2005 to 14%).

For each clinical outcome listed in the survey—A1C, body mass index, blood pressure, microalbumin, total cholesterol, triglycerides, and weight—collection and reporting decreased virtually every year. With statistically significant declines noted for 2008 data versus 2005 data (and often versus other years), a downward trend becomes apparent in clinical outcomes activity. However, 3 additional statistically significant results confound the ability to fully assess this area, as follows: (1) more respondents selected “other” in 2008 versus 2005 (10% vs 6%), (2) fewer respondents answered “none collected” in 2008 versus 2005 (1% vs 10%), and (3) more respondents answered “don’t know” in 2008 versus 2005 (5% vs 2%).

Data collection regarding 2 process measures—eye and foot examinations—also has decreased almost every year, showing statistically significant changes when comparing 2008 and 2005 results. However, as with clinical outcomes, clarification is needed because significantly more respondents selected “other” in 2008 (8%) than in 2005 (2%). Moreover, the selections of “none collected” decreased and “don’t know” increased significantly during this period.

Far more accreditation agencies and diabetes advisory groups received outcome reports in 2008 than in previous years. Nearly three-quarters (72%) of outcome reports went to diabetes advisory groups in 2008 versus an average of 39% in previous years, and 60% went to accreditation agencies in 2008 versus an average of 34% in previous years.

Quality/performance improvement efforts. A spike occurred in 2008 compared with previous years regarding participation in quality or performance improvement activities. This increase differed significantly from 2005 results, growing from 74% to 88%.

Adoption of the CCM. Because comparative data on the CCM can be obtained for only 2007 and 2008, it is too early to assess for trends. However, some statistically significant differences emerged between these 2 years. More respondents said they were familiar with the MacColl Institute’s CCM in 2008, and its application increased in self-management support and within the

health system. In addition, more respondents selected “other” when asked to identify the areas of their practice in which they used the MacColl CCM, and an increased number of respondents said they used a different CCM.

Limitations of the Study

The accuracy of the NPS in describing the practice of diabetes education in the United States may be limited by the constrained scope of the survey sampling. Because the sampling comprises only AADE members, it does not include the entire universe of diabetes education, which has been estimated (but not validated) to include up to 30 000 individuals.²

Conclusions and Recommendations

Many of the findings of the 2008 National Diabetes Education Practice Survey concur with those from previous surveys, serving to validate the general description of diabetes education practice. Moreover, the 2008 NPS has allowed the comparison of findings from 4 consecutive years, thereby uncovering some significant changes and emerging trends.

NPS data indicate that most diabetes education programs are well established, having been in place for more than 5 years, and are recognized by the ADA Education Recognition Program, Indian Health Service, or a state agency. Diabetes education takes place primarily in hospital-based settings and serves patient populations characteristic of the incidence of diabetes occurring in the general population. Nearly three-quarters of diabetes education programs are comprehensive.

Each year, NPS results have shown wide ranges in types of program services, instructional methods, length of program delivery, number of patients and visits, and team functions. This variability underscores the need to standardize interventions and provide guidelines for practice as a means to more consistently connect the provider and patient with the right service.²

Most patients succeed in completing their diabetes education program, and an overwhelming majority of educators elicit feedback via patient satisfaction surveys. Educators also are involved in measuring outcomes, with about 50% of diabetes education programs collecting and reporting at least some type of behavioral and clinical outcome and more than 75% programs assessing

learning outcomes. However, the collection of clinical outcomes is declining, and at present, program outcomes are not reported consistently on a national level. Clearly, evidence on the outcomes of diabetes education is crucial to elucidate its contribution to diabetes care. Therefore, educators are urged to increase their measurement and reporting of program outcomes.

Fiscal soundness is an important issue, as revealed by findings that diabetes education programs are evenly divided between those that operate at a financial deficit and those that are either revenue/cost neutral or profitable. Of course, many programs are not revenue generating because they do not charge for their services. Nevertheless, the practice of diabetes education would be well served by adopting business models that can improve the economic viability of programs.

The diabetes health community is beginning to embrace the CCM, a framework linking health care systems, the community, and persons with diabetes. The CCM is a useful way to address the different influences on health behavior change and to improve chronic illness care.⁷ However, with only 30% of diabetes educators familiar with the MacColl Institute's CCM, AADE should increase efforts to educate members on the benefits of this model as well as urge the diabetes health community to work toward adopting the CCM.

One enhancement recommended for future NPS instruments is the addition of questions assessing whether AADE members are practicing evidence-based diabetes care. In addition, a selection of either "Hispanic" or "Mexican

American/Latino American" could be incorporated into the question on racial distribution because this population is at high risk for type 2 diabetes. It is also recommended that AADE consider conducting the NPS biannually rather than annually, given that a solid foundation of data has now been obtained and to avoid survey fatigue among members. Regardless of whether the survey remains annual or becomes biannual, the NPS is likely to continue to help AADE reach its goals. Based on the data captured and trends identified through this survey, the association can apply insights to address members' needs and improve the practice of diabetes education.

References

1. Zrebiec J. A national study of the certified diabetes educator: report on a job analysis conducted by the National Certification Board for Diabetes Educators. *Diabetes Spectrum*. 2005;18:181-185.
2. Peebles M, Austin MM. Toward describing practice. The AADE National Diabetes Education Practice Survey: diabetes education in the United States—who, what, where, and how? *Diabetes Educ*. 2007;33:424-433.
3. American Association of Diabetes Educators. The scope of practice, standards of practice, and standards of professional performance for diabetes educators. *Diabetes Educ*. 2005;31:487-512.
4. Donebedian A. *The Definition of Quality and Approaches to Its Assessment*. Ann Arbor, MI: Health Administration Press; 1980.
5. AADE Position Statement. AADE7™ self-care behaviors. *Diabetes Educ*. 2008;34:445-449.
6. AADE Position Statement. Diabetes educators: implementing the chronic care model. *Diabetes Educ*. 2008;34:451-455.
7. Wagner EH, Austin BT, Davis C, Hindmarsh M, Schaeer J, Bonomi A. Improving chronic illness care: translating evidence into action. *Health Aff*. 2001;20:64-78.

Pages 971, 975-976, 979, 981, and 984 were intentionally removed from this PDF.
In the print version, these pages were advertisements.